Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

November 6, 2025

8:45–11:45

- This exam consists of Exercises **1–9 for 202001172** and **1–11 for 202001171**.
- You must **motivate** and **justify your answers clearly**. Answers without explanation will not receive credit.
- The use of calculators or any other electronic devices is not permitted.
- A formula sheet is attached to this exam. No other course materials are allowed.

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1. Suppose that both f and its derivative f' are continuous, and that

$$\lim_{x\to 2} \frac{f(x)-5}{x-2} = \frac{1}{2}.$$

Using the properties of limits and L'Hôpital's rule, find:

(a) $\lim_{x\to 2} f(x)$; (b) $\lim_{x\to 2} \left(\frac{x-2}{f(x)-5}\right)^2$; (c) $f'(2)$.

2. The functions f and g, together with their derivatives, satisfy the following values:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	1	3	2	-1
3	3	1	π	4

Find the derivatives with respect to x of the following expressions, evaluated at the indicated x values:

(a) $\dfrac{1}{f(x)g(x)}$ at $x=3$; (b) $f\big(g(x)-2e^{x-1}\big)$ at $x=1$.

3. Evaluate the following integrals:

(a) $\displaystyle\int \frac{\sin(\sqrt{x}+1)}{\sqrt{x}}\,dx$; (b) $\displaystyle\int_1^2 x^4 \ln(x)\,dx$.

4. Consider the triangle with vertices at the points:

$$P_1 = (1,0,2), \qquad P_2 = (3,2,0), \qquad P_3 = (0,4,1).$$

(a) Find an equation of the plane that contains the triangle and compute the area of the triangle.

(b) Find parametric equations of the line that passes through P_1 and is perpendicular to the plane from part (a). Verify that this line is perpendicular to the plane by showing that its direction ("velocity") vector is orthogonal to the vectors $\overrightarrow{P_1P_2}$ and $\overrightarrow{P_1P_3}$ which lie in the plane.

5. The polar equation
$$r = \frac{2}{1+\sin\theta}, \qquad (\sin\theta \neq -1)$$
can be expressed in Cartesian coordinates. Convert it to Cartesian form and identify the corresponding curve.

6. Show that the limit
$$\lim_{(x,y)\to(0,0)} \frac{2x}{x^2+x+y^2}$$
does **not** exist.

7. Let
$$z = \sin(xy) + x\sin y, \quad x = u^2+v^2, \quad y = uv.$$
Use the chain rule to find $\dfrac{\partial z}{\partial u}$ and $\dfrac{\partial z}{\partial v}$ at the point $(u,v) = (1,0)$.

8. Consider the function
$$f(x,y) = x^4 + y^4 - 4xy.$$
   (a) Find all critical points of f.
   (b) Determine the absolute minimum of f on the line segment defined by $x = 1$ and $-2 \leq y \leq 2$.

9. Solve the differential equation
$$x^2\frac{dy}{dx} + 3xy = x^4, \quad y(1) = 2, \quad \text{for} \quad x > 1.$$

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10. Find all solutions of the equation
$$x^4 + 7x^2 + 49 = 0,$$
and express all roots in Cartesian form $a + bi$.

11. Find the unique solution of the following second-order initial value problem:
$$y'' + 4y' + 13y = 0, \quad y(0) = 2, \quad y'(0) = -3.$$

Point distribution:

Exercise	1			2		3		4		5	6	7	8		9	10	11	Total
	a	b	c	a	b	a	b	a	b				a	b				
202001171	1	2	1	2	2	2	2	2	2	3	4	4	2	3	4	5	4	45
202001172	1	2	1	2	2	2	2	2	2	3	4	4	2	3	4	–	–	36

Exam grade (202001171): $1 + (\text{obtained points}) \times \frac{1}{5}$

Exam grade (202001172): $1 + (\text{obtained points}) \times \frac{1}{4}$

Solutions for Calculus A Exam

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1. (a) Using the product property of limits:

$$\left(\lim_{x\to 2}\frac{f(x)-5}{x-2}\right)\underbrace{\left(\lim_{x\to 2}(x-2)\right)}_{=0}=\lim_{x\to 2}\big(f(x)-5\big)=\lim_{x\to 2}f(x)-5=0.$$

Hence, $\boxed{\lim_{x\to 2}f(x)=5}$.

(b)

$$\lim_{x\to 2}\left(\frac{x-2}{f(x)-5}\right)^2=\left(\frac{1}{\lim_{x\to 2}\frac{f(x)-5}{x-2}}\right)^2=\left(\frac{1}{1/2}\right)^2=\boxed{4}.$$

(c) The limit $\lim_{x\to 2}\dfrac{f(x)-5}{x-2}$ is of the form $0/0$. Applying L'Hôpital's rule:

$$\lim_{x\to 2}\frac{f(x)-5}{x-2}=\lim_{x\to 2}f'(x)=f'(2)=\boxed{\frac{1}{2}}.$$

2. (a)

$$\frac{d}{dx}\left(\frac{1}{f(x)g(x)}\right)=-\frac{f'(x)g(x)+f(x)g'(x)}{(f(x)g(x))^2}.$$

At $x=3$:

$$-\frac{\pi(1)+3(4)}{(3\cdot 1)^2}=\boxed{-\frac{\pi+12}{9}}.$$

(b) Using the chain rule:

$$\frac{d}{dx}f\big(g(x)-2e^{x-1}\big)=f'\big(g(x)-2e^{x-1}\big)\big(g'(x)-2e^{x-1}\big).$$

At $x=1$:

$$f'(3-2)(-1-2)=f'(1)(-3)=2(-3)=\boxed{-6}.$$

3. (a) Let $u=\sqrt{x}+1$, so $\dfrac{du}{dx}=\dfrac{1}{2\sqrt{x}}$. Then

$$\int\frac{\sin(\sqrt{x}+1)}{\sqrt{x}}\,dx=2\int\sin(u)\,du=-2\cos(u)+C=\boxed{-2\cos(\sqrt{x}+1)+C}.$$

(b) Integration by parts with $u=\ln x$, $\dfrac{dv}{dx}=x^4$ gives

$$\int_1^2 x^4\ln x\,dx=\frac{x^5\ln x}{5}\Big|_1^2-\frac{1}{5}\int_1^2 x^4\,dx=\frac{32\ln 2}{5}-\frac{1}{5}\left[\frac{x^5}{5}\right]_1^2=\frac{32\ln 2}{5}-\frac{31}{25}=\boxed{\frac{160\ln 2-31}{25}}.$$

4. (a) Vectors in the plane:

$$\overrightarrow{P_1P_2} = P_2 - P_1 = (2, 2, -2), \qquad \overrightarrow{P_1P_3} = P_3 - P_1 = (-1, 4, -1).$$

A normal vector to the plane is the cross product:

$$\mathbf{n} = \overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & -2 \\ -1 & 4 & -1 \end{vmatrix} = (6, 4, 10).$$

Hence, an equation of the plane through P_1 is

$$6(x-1) + 4(y-0) + 10(z-2) = 0,$$

which simplifies to

$$\boxed{3x + 2y + 5z - 13 = 0.}$$

The area of the triangle is half the magnitude of the cross product:

$$A = \frac{1}{2}\|\mathbf{n}\| = \frac{1}{2}\sqrt{6^2 + 4^2 + 10^2} = \frac{1}{2}\sqrt{152} = \boxed{\sqrt{38}}.$$

(b) A line through P_1 with direction vector $\mathbf{n}$ is

$$\mathbf{r}(t) = P_1 + t\mathbf{n} = (1, 0, 2) + t(6, 4, 10),$$

so the parametric equations are

$$\boxed{x = 1 + 6t, \qquad y = 4t, \qquad z = 2 + 10t, \quad t \in \mathbb{R}.}$$

Verification of orthogonality:

$$\mathbf{n} \cdot \overrightarrow{P_1P_2} = 6(2) + 4(2) + 10(-2) = 12 + 8 - 20 = 0,$$

$$\mathbf{n} \cdot \overrightarrow{P_1P_3} = 6(-1) + 4(4) + 10(-1) = -6 + 16 - 10 = 0.$$

5.

$$r(1 + \sin\theta) = 2 \implies \sqrt{x^2 + y^2} + y = 2 \implies x^2 = 4 - 4y.$$

Thus $\boxed{y = 1 - \frac{x^2}{4}}$, a downward-opening parabola with zeros at $x = \pm 2$.

6. Along $y = 0$: $\lim\limits_{x \to 0} \dfrac{2x}{x^2 + x} = 2$. Along $x = 0$: $\lim\limits_{y \to 0} \dfrac{0}{y^2} = 0$. Since these limits differ, the overall limit does not exist.

7.

$$\frac{\partial z}{\partial x} = y\cos(xy) + \sin y, \qquad \frac{\partial z}{\partial y} = x\cos(xy) + x\cos y.$$

At $(u, v) = (1, 0)$: $x = 1$, $y = 0$, so $\dfrac{\partial z}{\partial x} = 0$, $\dfrac{\partial z}{\partial y} = 2$. We also have the partial derivatives of x and y:

$$\frac{\partial x}{\partial u} = 2u, \quad \frac{\partial x}{\partial v} = 2v, \quad \frac{\partial y}{\partial u} = v, \quad \frac{\partial y}{\partial v} = u.$$

At $(u, v) = (1, 0)$, these are $\frac{\partial x}{\partial u} = 2$, $\frac{\partial x}{\partial v} = 0$, $\frac{\partial y}{\partial u} = 0$, $\frac{\partial y}{\partial v} = 1$. By the chain rule:

$$\boxed{\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} = (0)(2) + (2)(0) = 0}.$$

$$\boxed{\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v} = (0)(0) + (2)(1) = 2}.$$

8. (a) Making the gradient zero:

$$f_x(x,y) = 4x^3 - 4y = 0 \implies y = x^3, \quad f_y(x,y) = 4y^3 - 4x = 0 \implies x = y^3.$$

Hence $x = x^9 \implies x = 0, \pm 1$. Critical points: $\boxed{(0,0),\ (1,1),\ \text{and } (-1,-1)}$.

(b) On the line segment defined by $x = 1$ and $-2 \le y \le 2$, we have

$$f(1,y) = 1 + y^4 - 4y.$$

Let $g(y) = f(1,y)$. Then

$$g'(y) = 4y^3 - 4 = 4(y^3 - 1).$$

Setting $g'(y) = 0$ gives $y^3 = 1 \Rightarrow y = 1$. We now evaluate g at the critical point and at the endpoints of the interval:

$$g(-2) = 1 + 16 - 4(-2) = 25, \qquad g(1) = 1 + 1 - 4 = -2, \qquad g(2) = 1 + 16 - 8 = 9.$$

Hence, the absolute minimum on the segment occurs at $y = 1$, with

$$\boxed{f(1,1) = -2}.$$

9. The equation can be written as

$$\frac{dy}{dx} + \frac{3}{x}y = x^2.$$

Integrating factor: $v(x) = x^3$.

$$(x^3 y)' = x^5 \implies x^3 y = \frac{x^6}{6} + C \implies y = \frac{x^3}{6} + \frac{C}{x^3}.$$

Using $y(1) = 2$ gives $C = \frac{11}{6}$. Hence

$$\boxed{y = \frac{x^3}{6} + \frac{11}{6x^3}}.$$

10. Let $u = x^2$. The equation becomes $u^2 + 7u + 49 = 0$. Using the quadratic formula:

$$u = \frac{-7 \pm \sqrt{49 - 4(49)}}{2} = \frac{-7 \pm \sqrt{-3 \cdot 49}}{2} = \frac{-7 \pm 7i\sqrt{3}}{2}.$$

In polar form, this is $u = 7e^{i(2\pi/3)}$ and $u = 7e^{i(4\pi/3)}$. Now we find the square roots of u to get x:

$$x = \sqrt{7}e^{i\pi/3},\ \sqrt{7}e^{i(2\pi/3)},\ \sqrt{7}e^{i(4\pi/3)},\ \sqrt{7}e^{i(5\pi/3)}.$$

In Cartesian form, the four roots are:

$$\boxed{x_1 = \frac{\sqrt{7}}{2} + i\frac{\sqrt{21}}{2}, \quad x_2 = -\frac{\sqrt{7}}{2} + i\frac{\sqrt{21}}{2}, \quad x_3 = -\frac{\sqrt{7}}{2} - i\frac{\sqrt{21}}{2}, \quad x_4 = \frac{\sqrt{7}}{2} - i\frac{\sqrt{21}}{2}}.$$

11. Characteristic equation: $r^2 + 4r + 13 = 0 \implies r = -2 \pm 3i$.

$$y(t) = e^{-2t}\big(A\cos 3t + B\sin 3t\big).$$

From $y(0) = 2$, $A = 2$. Then

$$y'(t) = e^{-2t}\big[-2(A\cos 3t + B\sin 3t) - 3A\sin 3t + 3B\cos 3t\big].$$

At $t = 0$: $y'(0) = -2A + 3B = -3 \implies B = \frac{1}{3}$. Thus

$$\boxed{y(t) = e^{-2t}\left(2\cos(3t) + \frac{1}{3}\sin(3t)\right)}.$$